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LA Assignment 2 - Week 1

Problem 1 & 2

Problem 1

(a) For $u = (r_1, g_1, b_1)$, $v = (r_2, g_2, b_2)$, w and scalars k, m, every axiom reduces to the corresponding property of real numbers, applied in each coordinate:

1) Closure under $\oplus$: the sum of two triplets of reals is a triple of reals.
2) Closure under $\odot$: $k(r, g, b)$ is a triple of reals
3) $u \oplus v = v \oplus v$, since real addition is commutative
4) $(u \oplus v) \oplus w = u \oplus (v \oplus w)$, since real addition is associative.
5) <u>zero vector</u> : $0 = (0,0,0)$, and $u \oplus 0 = u$
6) Additive inverse : $-u = (-r, -g, -b)$, and $u \oplus (-u) = (0,0,0)$
7) $k \odot (u \oplus v) = k \odot u \oplus k \odot v$, since $k(a+b)$ $= ka + kb$.
8) $(k+m) \odot u = k \odot u \oplus m \odot u$
9) $k \odot (m \odot u) = (km) \odot u$
10) $1 \odot u = u$

All ten axioms hold so V is a vector space.

(b) Bounded operations. V is not a vector space.
Several axioms fail:

1- Associativity of $\oplus$. Take $a = (1,1,1)$, $b = (1,1,1)$, $c = (-1,-1,-1)$.
$(a \oplus b) \oplus = (1,1,1) \oplus C = (\min(0,1), \dots)$
$= (0,0,0)$, but $a \oplus (b \oplus c) = a \oplus (0,0,0) = (1,1,1)$.
These differ.

2- Zero vector. We would need e with $x \oplus e = x$ for every $x \in R^3$. Take $x = (2,2,2)$: every output of $\oplus$ has components ≤ 1, so $x \oplus e \neq x$ for any e. No vector exists.

3- Additive inverses. With no zero vector, inverses are undefined. Concretely, $(1,1,1) \oplus v = (0,0,0)$ needs $v = (-1,-1,-1) \oplus v = (0,0,0)$ while $(2,2,2) \oplus v'$ can never reach 0 for any v' that works for $(1,1,1)$ (and inverses for vectors with components >1 don't exist either).

4- Identity for $\odot$. $1 \odot (2,2,2) = (1,1,1)$, $k=2$, $m=-1$.
LHS $= 1 \odot u = (1,1,1)$. RHS $= (1,1,1) \oplus (-1,-1,-1)$
$= (0,0,0)$

5- Compatibility $k \odot (m \odot u) = (km) \odot u$. Take $u = (1,1,1)$, $m = 2$, ~~$k=1$~~ $k = -1$. $k \odot (m \odot u) = -1 \odot (1,1,1) = (-1,-1,-1)$, but $(km) \odot u = (-2,-2,-2)$.

Problem 2

(b) $W_2 = \{x : x_1 + x_2 + x_3 + x_4 = 1\}$: not a subspace

1- The zero vector $(0,0,0,0)$ has sum $0 \neq 1$, ~~so 0 ∈~~ so $0 \notin W_2$.

2- Closure fails : $u = (1,0,0,0)$ and $v = (0,1,0,0)$ are in W_2, but $u+v = (1,1,0,0)$ has sum $2 \neq 1$. So $u+v \notin W_2$.

(c) $W_3 = \{(x_1, x_2, x_3, x_4) : x_1 = 2x_3 \text{ and } x_2 = x_4\}$

(i) Subspace, check the three condition

1) Nonempty : for $(0,0,0,0)$, $x_1 = 0 = 2 \cdot 0$ and $x_2 = 0 = x_4$, so $0 \in W_3$.

2) Addition : let $u = (u_1, u_2, u_3, u_4)$ and $v = (v_1, v_2, v_3, v_4)$ be in W_3, so $u_1 = 2u_3$, $u_2 = 2u_3$, $u_2 = u_4$, $v_1 = 2v_3$, $v_2 = v_4$. Then

- $(u+v) = u_1 + v_1 = 2u_3 + 2v_3 = 2(u_3 + v_3) = 2(u+v)_3$
- $(u+v)_2 = u_2 + v_2 = u_4 + v_4 = (u+v)_4$.

So $u+v \in W_3$.

3) Scalar multiplication : for $\in R$,

1- $(ku)_1 = ku_1 = k(2u_3) = 2(ku_3) = 2(kv)_3$

$(kv)_2 = kv_2 = kv_4 = (kv)_4$

So $kv \in W_3$.

All three condition hold, so W_3 is a subspace of R^4.

c)(ii) Not applicable, since W_3 is a subspace.

Extra (basis and dimensions): every element has the form $(2s, t, s, t) = s(2,0,1,0) + t(0,1,0,1)$, so $W_3 = \text{span}\{(2,0,1,0), (0,1,0,1)\}$ and $\dim W_3 = 2$.